

ANN MARY THOMAS

Data Analyst | Sustainability Researcher | AI & Machine Learning Specialist

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PROFESSIONAL SUMMARY

Data analyst and sustainability researcher with expertise in energy analytics, carbon forecasting, and AI-driven predictive modeling. Currently employed at WMG, University of Warwick, supporting advanced manufacturing SMEs through the government-funded STRATA project. Proven track record in developing deep learning models for electricity demand and CO₂ emissions forecasting, achieving 39.3% carbon footprint reduction through data-driven analysis. Skilled in translating complex data into actionable strategies for decarbonization and operational efficiency. Strong foundation in mechanical engineering combined with advanced data analytics capabilities.

TECHNICAL SKILLS

Data Science & AI: LSTM, RNN, Attention Mechanisms, Hybrid Deep Learning Models, Time-Series Forecasting, Predictive Modeling, Regression Analysis, Classification

Programming & Tools: Python (Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow), Power BI, SQL, Excel (Advanced), Git, Jupyter Notebook

Energy & Sustainability: Electricity Demand Modeling, CO₂ Emissions Forecasting, Decarbonization Strategies, Renewable Energy Integration, Carbon Intensity Metrics

Professional Skills: Analytical Thinking, Data Storytelling, Stakeholder Engagement, Research & Documentation, Problem Solving, Team Collaboration

PROFESSIONAL EXPERIENCE

Energy Consumption and Sustainability Analyst

WMG, University of Warwick | Dec 2025 – Present | Coventry, UK

Deployed to 3T Additive Manufacturing as part of the government-funded STRATA project, providing specialist analytical support for energy optimization and decarbonization strategies in additive manufacturing operations.

- Conduct comprehensive energy consumption analysis and develop roadmaps for carbon footprint reduction across manufacturing operations
- Build data-driven models and Power BI dashboards to monitor operational efficiency, energy metrics, and sustainability KPIs
- Provide analytical support for the STRATA project, focusing on sustainable manufacturing practices and process optimization
- Collaborate with engineering teams to implement evidence-based policies and improvement initiatives
- Perform scenario analysis and cost-benefit assessments to support strategic decision-making
- Create technical reports and presentations for stakeholders and funding bodies

Energy Consumption and Sustainability Analyst (Intern)

3T Additive Manufacturing | Sep 2025 – Dec 2025 | Newbury, UK

- Analyzed manufacturing process data to identify energy consumption patterns, bottlenecks, and optimization opportunities
- **Achieved 39.3% carbon footprint reduction through systematic data analysis and implementation of targeted energy-saving strategies**
- Defined and tracked key performance indicators (KPIs) for energy efficiency, operational costs, and CO₂ emissions
- Developed forecasting models and interactive Power BI dashboards to support real-time decision-making
- Created technical reports and presentations communicating analytical findings to stakeholders

Part-Time Research Assistant

Genesis Lab, London Metropolitan University | Feb 2025 – Present | London, UK

- Spearheading research on deep learning-based forecasting of electricity demand and CO₂ emissions addressing critical sustainability challenges

- Independently designed and implemented novel Hybrid LSTM+Attention architecture, achieving superior accuracy over traditional RNN and RBM+MLP approaches
- Conducting rigorous scenario-based analysis simulating 10-50% renewable energy penetration to quantify CO₂ reduction potential
- Performing regional benchmarking across California and Texas energy grids, providing actionable insights for policymakers
- Preparing research findings for academic publication in peer-reviewed journals

Junior Data Analyst

Navalt Solar and Electric Boats | Aug 2022 – Jan 2024

- Led analysis of operational data for 600+ ships from Mediterranean Shipping Company, processing 100,000+ records
- Developed Python automation scripts reducing data preparation time by 75% and improving accuracy
- Built predictive models to evaluate fuel efficiency and estimate CO₂ emissions using sea state and engine performance variables
- Researched Carbon Intensity Indicator metrics using IMO guidelines, expanding domain knowledge in maritime sustainability

EDUCATION

MSc in Data Analytics

London Metropolitan University | 2024 – 2025

B.Tech in Mechanical Automobile Engineering

SCT College of Engineering | 2017 – 2021

KEY PROJECTS

Energy Forecasting with Deep Learning

- Implemented LSTM, RNN, and RBM+NN models to forecast electricity demand and CO₂ emissions in U.S. power sector
- Achieved 5% improvement in forecast accuracy with optimized LSTM model; findings prepared for academic publication

Vehicle Emissions Dashboard

- Developed Power BI dashboard analyzing CO₂ emissions across vehicle types using EU environmental dataset, demonstrating engine size drives 70% of emission variance

Graduate Outcomes in Education

- Applied Random Forest and K-means clustering to UK higher education data, achieving 45% classification accuracy and identifying four distinct graduate profiles

CERTIFICATIONS

- Excel for Data Analytics – Simplilearn
- Statistics for Data Science – Simplilearn

ACHIEVEMENTS

- **Achieved 39.3% carbon footprint reduction through data-driven sustainability initiatives at 3T Additive Manufacturing**
- Research on energy demand and decarbonization selected for publication under Genesis Lab
- Independently designed LSTM+Attention hybrid model architecture for critical infrastructure forecasting
- Student volunteer for Frontiers of Intelligent Computing: Theory and Applications (FICTA'25) conference
- Reduced data processing time by 75% through Python automation at Navalt Solar and Electric Boats